

# Millet stem borer

*Coniesta ignefusalis*

|  | Prevention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Monitoring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Direct Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|  | <ul style="list-style-type: none"> <li>• Use resistant varieties if locally available.</li> <li>• Plant early, with the first rains, to avoid a heavy infestation of the stem borer.</li> <li>• Intercrop millet with non-host crops, such as cowpeas.</li> <li>• The 'push-pull' method is effective against stem borers:</li> <li>• <i>Desmodium</i> is intercropped with millet and acts as a repellent that 'pushes' the pest away from the millet.</li> <li>• Around the border of the field a trap crop, such as Napier or Sudan grass, is planted to attract or 'pull' the stem borer moths away from the millet.</li> <li>• The moths lay eggs in the trap crops; when the larvae bore into the Napier grass, the plant produces a sticky glue-like substance that traps the larvae and they die.</li> <li>• Plant strips of flowering plants in or around the field to attract and conserve the natural enemies such as ants and parasitic wasps (<i>Telenomus busseolae</i>, <i>Syzeuctus</i> spp. or <i>Cotesia sesamiae</i>).</li> <li>• Rotate millet with a non-host crop, such as a legume, to break the pest's life cycle.</li> </ul> | <ul style="list-style-type: none"> <li>• Additional relevant crops: sorghum (especially susceptible when intercropped with maize), maize and sugarcane.</li> <li>• Monitor from germination to harvest for insects and symptoms.</li> <li>• Insect:</li> <li>• Eggs: 1 mm long, elliptical and yellowish-white in colour. They are laid in batches of 20-25 between the leaf sheaths and stem or on lower leaf blades.</li> <li>• Larvae: about 20 mm long when fully grown, have a reddish-brown head and a creamy white body, with distinct black spots along the body during the wet season.</li> <li>• Pupa: up to 15 mm long, yellowish to reddish brown with thorn-like spikes on the abdominal segments and normally found within the stem.</li> <li>• Adult moths: have a wing span of about 8 to 15 mm, golden-brown forewings and white silky hind wings.</li> <li>• Damage:</li> <li>• The larvae attack the growing points and leaves of millet, and bore into the stem causing 'dead hearts'.</li> <li>• Small plants attacked may be thoroughly riddle with caterpillars and soon collapse.</li> <li>• In larger plants, external symptoms show two to three weeks after stems have been infested.</li> <li>• Use pheromone bait traps to monitor adult stem borers.</li> <li>• Take control measures as soon as the stem borers are detected in the fields.</li> </ul> | <ul style="list-style-type: none"> <li>• Spray neem products early in the season before the larvae bore into the stem of the plant; apply in the evenings.</li> <li>• Burn or bury at 5 cm crop residues after harvest as the stem borer larvae can survive until the next rainy season in the stems and stubble.</li> <li>• Place locally made pheromone bait traps in and around the field to disrupt the mating cycle of the stem borers and reduce the populations. The traps are particularly beneficial if placed along fences and granaries, or other places the stem borer may hide.</li> <li>• If the stems are to be used for building structures, they should be partially burned to kill the larvae inside the stem.</li> </ul> |

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.